| | | |
|---|---|---|
| | | Since molecular KE is proportional to temperature, the molecular KE changes 4 times too. |
| **14** | **C** | $KE = \frac{1}{2}m\omega^2(x_0^2 - x^2)$<br>$PE = \frac{1}{2}m\omega^2 x^2$<br>$\frac{KE}{PE} = \frac{x_0^2 - x^2}{x^2} = \frac{x_0^2 - \left(\frac{1}{2}x_0\right)^2}{\left(\frac{1}{2}x_0\right)^2} = \frac{4-1}{1} = 3$ |
| **15** | **C** | At resonance, $x_0$ = 16.4 cm and $f$ = 1.5 Hz (read from graph) |